

Beyond Convergence: A Lightweight Human-in-the-Loop Semantic Conflict Triage Framework for Offline-First CRDT-Based Collaborative Text Editing

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Abstract- Offline-first collaborative editors can save locally made changes and convergence will eventually reconcile replicated state. Convergence, however, does not guarantee that changes made concurrently by different users are semantically correct. This paper considers an integration of offline-first CRDT-based collaborative rich-text editing, analysis of concurrent edits, and reconciliation with a human in the loop. This paper does not attempt to validate the novelty of semantic conflict detection. Concurrent transactions in Yjs are filtered using causal and structural metadata, low-cost lexical features, Sentence-BERT, and bidirectional natural language inference. Only unresolved candidate pairs are escalated to a large language model and, if necessary, to a reviewer. The framework defines edit pairs as duplicate, complementary, contradictory, or independent, and is mapped to review actions of the same name. Each action is designed to be as non-invasive as possible. A Wikipedia-based corpus of 300 revisions is used to construct semantic-intent benchmarks. Initial evaluation of the intent annotations revealed 246 intent annotations, 215 coded revisions, 26 multi-intent revisions, and nine review-tier inconsistencies caused by code-zero handling. To evaluate the intended system, controlled offline/reconnection replay is specified to examine macro-F1, contradiction recall, automated coverage, review workload, latency, and model-call rate, while ensuring CRDT merge semantics are not altered. It is intended that the separation of synchronization correctness from content-level editorial judgment be maintained during the replay. Automated evaluation of 300 two-annotator labeled edit pairs (Cohen's $\kappa = 0.800$) at full-cascade shows 80.7% LLM-avoidance coverage and 87.3% recall of contradictions in safety-critical class.

Keywords— CRDT, offline-first, collaborative editing, semantic conflict detection, human-in-the-loop, natural language inference, Yjs, LLM.

I. INTRODUCTION

Collaboration based on an offline-first model fundamentally alters how we think about the failure mode of joint document editing. Rather than assume that lack of connectivity is a rare event and block further editing, each user continues to edit a local copy of the document and reconciles the editing conflicts upon connectivity. The literature on operational transformation and CRDTs provides solutions to the problem of concurrent editing and ensures that a deterministic or eventual consistent convergence across the copies is achieved. Local-first systems take this one step further and treat local copies as the normal case, as opposed to a transient cache. Perfect convergence of two replicas, however, is not the only correctness criterion for collaborative editing of prose. There are cases where the merged document contains redundant, inconsistent, misleading, or editorially incompatible statements.

Let us consider a case where the two users disconnect after reading the sentence "The service retains logs for seven days." In the interim,

one user edits the sentence to read "The service retains logs for thirty days" after consulting a more recent policy, while the other user edits the document to read "Logs are deleted after seven days" from an older policy. A CRDT would preserve the two edits and integrate them at a later time (4), (5). There are also CRDTs for rich text which also capture formatting intent and preserve the order of edits (7)-(9). There are, however, possible conflicting claims in the document that are semantically editorial issues rather than a failure of the reconciled state. There are also cases where the two users restate the same fact resulting in redundant text, or where the two users edit compatible information which should be proximal in the document. This paper will examine post-convergence semantic triage instead of developing a new replication algorithm. For the time being, the responsibility for local editing, causal update exchange, and convergence of the Yjs document will not change. An observing layer will only examine candidate spans associated with concurrent or overlapping offline edits. Recent advancements in semantic conflict resolution for collaborative data structures focus on modeling application-level dependencies [10], while previous research on design documents resolved semantic conflicts through domain ontologies [11]. This paper does not intend to discredit the effort that has been dedicated to the detection of semantic conflicts in CRDT-related systems. The focus of this paper is on the combination of offline-first, CRDT-based, collaborative rich-text editing, analysis of concurrently authored observations, and semantic reconciliation.

The evaluation will be structured around five research questions. The first, RQ1, will focus on the four-class model and will evaluate its ability to distinguish between duplicate, complementary, contradictory, and independent edit pairs. RQ2 will consider selective routing and evaluate if it maintains a high level of contradiction recall. As preceding work has shown, in editing, it is more detrimental to miss a contradiction than to unnecessarily prompt an editor to review his or her work [13], [14], [16]. RQ3 will evaluate the impact of semantic triage on the reviewer workload. RQ4 will evaluate the incremental cost of semantic triage in terms of increased latency, large-model calls, and processing time. Finally, RQ5 will focus on the semantic intent and review risk derived from revision analysis [12], and consider human-AI revision studies to frame reviewer-centered evaluation [17], [18].

This integration considers only a small number of specific examples. In this context, the main ideas are a four-class relation taxonomy constructed from conservative examples; a cost-sensitive approach that combines structural pruning, Sentence-BERT, bidirectional natural language inference, large-model reasoning, and human verification; a risk-sensitive annotation procedure that constructs validated concurrent-edit pairs from Wikipedia revision data; and a reproducible approach to offline/reconnection Yjs. The framework defines coverage-risk analysis, class-specific metrics, latency and call-rate measurements, and ablations to determine if selective escalation is useful compared to rule-only, embedding-only, inference-only, and large-model-only baselines [12]-[20]. The core

idea is to separate concerns: replication correctness is guaranteed, and the concern of semantic uncertainty becomes visible and can be addressed. This design ensures that collaborators retain control over what interventions are made, and that interventions are both clearly defined and traceable.

II. RELATED WORK AND RESEARCH GAP

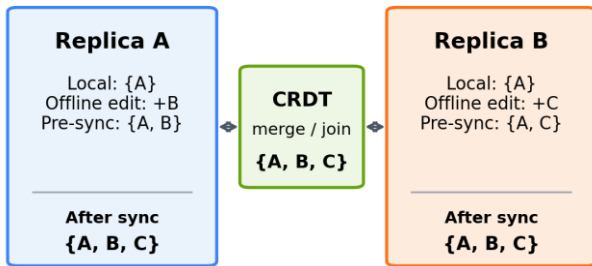
A. CRDT and Local-First Foundations, Rich-Text Intent, and Post-Convergence Limitations

Early studies on collaborative editing focused on transforming or commuting simultaneously performed operations to allow users to edit the same document without having to perform edits serially. Sun and Ellis studied operational transformation algorithms and the consistency problems caused by concurrent edits. [1] Prego et al. later developed a commutative replicated approach to cooperative editing [2], and Shapiro et al. formalized CRDTs, replicated data structures whose conflicts are resolved without centralized control, and whose operations or states can be combined to ensure that all replicas converge. [3] Kleppmann and Beresford built on this work to propose a way to merge replicated state of structured JSON to maintain concurrent edits. [4] In this line of work, the main concerns are guarantees of causality, update preservation, deterministic integration, and convergence.

Local-first software provides the deployment context where these guarantees are critically needed. Local-first applications provide a usable local copy to each user, allow users to perform edits while disconnected, and synchronize their edits upon reconnection. [5] While this model improves availability and ownership of the data, merge conflicts cannot be used to guarantee the correctness of the data. Schiefer et al. showed that some application invariants cannot be merged and would require explicit forking or reconciliation. [6] This is important to collaborative writing systems. A system may be able to combine edits, but the content may not be valid to accept.

Recent improvements to rich text focus on preserving structural intent. Peritext encodes formatting spans and editing operations using a particular kind of CRDT to facilitate collaborative rich text editing. Eg-walker focuses on efficient merging across long editing histories and offline branches. Fugue resolves undesirable interleaving of concurrent edits and maintains the convergence of CRDTs. These additional features and improvements remove some structural inconsistencies, but by themselves, do not guarantee that two pieces of natural language text align in meaning after they have been synchronized.

Fig. 1. State-Based CRDT Convergence Example



Source: Adapted from Julia y Carlos, "Diagrama CRDT basado en estado," Wikimedia Commons, 20 May 2021, CC0 1.0.

https://commons.wikimedia.org/wiki/File:Diagrama_CRDT_basado_en_estado.png
Explanation: The replicated-state example demonstrates that convergence is a synchronization property. Undesirable content can still converge, so a separate review layer at the content level is warranted.

A relevant systems-level reference is the Semantic Conflict Model, which describes some reconciliation needs by representing semantic inter-dependencies among collaboratively executed operations. This model of semantics encompasses inter-dependencies among operations or inter-dependencies among the application of operations, as opposed to inter-dependencies among pairs of text expressed in natural language. Therefore, this work is not the first to consider semantic conflicts in replicated systems. It considers content-semantic integration after synchronization, keeping CRDT

merge rules unchanged, and using NLP to identify passages that likely require manual editing, to integrate content-semantic synchronization.

B. Semantic Revision Analysis, NLI, and the Underexplored Integration Gap

Semantic analysis of revisions provides complementary techniques, and has typically been handled separately from offline-first replication. Gu et al. used ontologies and semantic knowledge related to WordNet to resolve conflicts in collaboratively edited design documents [11]. While this example shows meaning-aware reconstruction in practice, it employs a domain-specific representation, which is both dependent on the task and precedes contemporary local-first CRDTs, and does not make use of causal metadata from disconnected replicas. The setting proposes meaning reconstruction as a post-sync service, which is integrated with a rich text editor that is capable of comparing edit pairs based on CRDT metadata.

Yang et al. proposed a semantic categorization of edit intentions in Wikipedia revisions, such as fact updates, copy editing, wikification, elaboration, verifiability, and related groups [12]. These terms describe the intent behind a revision, but not how a pair of concurrent revisions are related to each other. For example, a marked fact update edit may be duplicated, supplementary, contradictory, or a recent edit of an independent fact. Intention is useful as a risk prior and sampling stratum, but is unfit to represent pairwise conflict resolution.

Natural language processing provides a solid basis for relation detection. The annotation of large amounts of text in the Stanford Natural Language Inference corpus provides labels of entailment, contradiction, and neutrality for pairs of sentences [13]. MultiNLI extends the scope of inference evaluation to different genres [14]. Semantic embeddings generated by Sentence-BERT provide an easy way to compute similarity and retrieve pairs of sentences [15]. For factual statements, the FEVER dataset provides an example of verification of factual statements along with evidence that supports or refutes the statements [16]. These datasets and models encourage using inexpensive semantic similarity checks to rule out a large number of cases before performing costly inferences or generative reasoning.

The focus of human-AI writing research is also on the interactive nature of revision decisions as opposed to predictive decisions. CoAuthor traces both human and language model inputs during the writing process [17]. Re3 models document revision processes at a broader level [18]. While RARR and SelfCheckGPT provide models to assist revision and self-checking, they also highlight the need to validate model reliability rather than accept generated rationales as correct [19,20]. The focus of this work is to fill a small, deliberately unmet need in the field: there is related work that covers the various components, but there has been relatively little direct assessment of integrating these components in offline-first, collaborative, rich-text editing with a focus on controlled reconciling review. This work combines CRDT metadata of concurrency, Semantic Analysis of concurrent observations with four-way Natural Language Relation Detection, along with controlled reconciliation during reviewed reconnects, and Human-in-the-Loop with measured review workload [7]-[12], [18]-[20].

III. PROPOSED FRAMEWORK AND ARCHITECTURE

A. Concurrent-Edit Candidate Generation and Lightweight Semantic Features

The semantic layer will connect to the editor at the boundaries of transactions in Yjs. For transactions that the observer knows were authored locally, it will store a replica identifier, causal metadata, the block or paragraph changed, the span that was changed, limited before-and-after context, the connectivity state, and the time of the next reconnect event. These records will be used for analysis. They will not affect the operations of Yjs or the order in which the CRDT integrates updates. This will allow us to maintain the local-first property, which states that an edit will be immediately visible on the

device where the edit was made and will be synchronized to the other devices through the ordinary mechanisms of Yjs [5] and [7]-[10]. Candidate generation will only occur after the system has identified edits that potentially interacted with each other. Two edits will be recognized as a semantic candidate if both edits occur simultaneously. i.e., there is no causal relationship between the edits. Edge cases are filtered using structural locality. This will also be the case when using language models [8]-[10].

For surviving candidates, the framework computes less expensive features next: character and token overlap, normalized edit distance, relative span position, block identity, changed-span distance, semantic-intention indicators, and a review-risk prior taken from the revision codebook [12]. Sentence-BERT encodes the edited propositions and surrounding context, and measures the cosine similarity as a proxy for semantic distance [15]. Extremely low similarity and distinct structural targets can justify making an independent decision, while extremely high similarity and lexical overlap can suggest a near-duplicate. Thresholds are set using validation data, as opposed to being determined using test data.

Pairs for which an interaction is still possible are further processed using natural language inference in both directions. The model gives Edit A conditional Edit B and Edit B conditional Edit A probabilities of entailment and contradiction, as well as a neutral probability [13], [14]. Scoring in both directions helps discern asymmetric entailment from equivalence and cases where one edit completes the other. High bidirectional entailment and high lexical or embedding similarity is best described as duplication. Asymmetric entailment paired with neutral-with-high semantic overlap to add novel information describes complementary. A high contradiction score in either direction is best described as contradictory. Neutral scores with low semantic and structural relatedness describes independent. Therefore, NLI labels are described as complementary with similarity and locality features rather than one-to-one mapped to the four classes. It is also possible that some fact-updates and verifiability cases contain evidence-check indicators based on factual verification [16]. This step produces a probability vector for the relations of duplicate, complementary, contradictory, and independent, along with a risk, overlap, and routing metadata for control of routing actions in subsequent routing steps.

B. Selective LLM Escalation and Human Verification

The framework relies on a cascade rather than routing every candidate to a large language model. Clear cases are handled through deterministic structural checks, lexical features, Sentence-BERT similarity, and bidirectional inference. Unresolved cases are evaluated to determine semantic uncertainty, contradiction probability, and review-risk. This design philosophy aims to provide a "lightweight" architecture that reduces the number of expensive model calls and reviewer interventions, while keeping contradiction recall within acceptable limits [15], [19], [20]. The operating thresholds are set on validation data and kept fixed for the final version of the testing.

When an escalation is necessary, the large model is given only local context. This includes the base passage, Edits A and B, causal and other metadata, and some inexpensive scoring. A prompt is given to the model to denote one relation, with evidence spans, and to provide a confidence estimation and suggested editing action. The schema supports an "uncertain" outcome, so the model is not required to give a certain answer when the evidence is weak. Evidence spans and confidence scores simplify logging and promote traceability, while giving explicit examples to the model, rather than relying on unexplained edits, justifies the need for retained evidence [18]-[20]. Routing policy is conservative. Unrelated or complementary changes that can be kept with confidence are permitted. An edit that is deemed to have a high confidence level may lead to a suggestion of non-destructive deduplication. However, the system will maintain both versions until there is a change to the routing policy or a reviewer makes a decision to merge the changes. Changes that are contradictory are always sent for manual resolution. Changes that are of a high confidence level and are deemed to be of a high risk are also sent for manual resolution, even if the top predicted class is not

a contradiction [6], [10], [18]. This is because accepting incompatible changes creates a greater risk.

The review interface provides the reviewer with the base context, locally and causally associated metadata, class probabilities, model evidence, and Span, as well as a justification for the edits. The reviewer can keep both changes, keep either change A or change B, or reconcile the changes. The system does not automatically change collaborator texts. Reviews are logged as labels and feedback, which will be used for further calibration and assessment against human reviews [17]-[20]. The same event record is kept of the model, prompt, thresholds, timing, and routing policy. This ensures that there is an auditable trace of simultaneous operations and interventions on the data. This policy ensures that reviewer effort is an explicit system resource.

IV. SEMANTIC RECONCILIATION MODEL

A. Four-Class Relation Taxonomy and Action Policy

The reconciliation model differentiates between semantic compatibility and synchronization success by assigning each concurrent-edit pair to one of four operational relations. In the case of a duplicate pair, it is assumed that the pair contains the same (or very similar) information. Higher confidence duplicates may be allowed to pass with a suggestion for consolidation, but lower confidence duplicates are not deleted automatically, as small changes in wording may represent a different viewpoint or evidence. As such, lower confidence duplicates are left for reviewer choice [13], [15].

In the case of a complementary pair, it is assumed that the pair contains information that improves the same local topic, with the two edits contributing compatible information that can be retained together. One edit may contain a justification for a claim, and the other pair edit may contain evidence for that claim. It is assumed that both edits should be retained, and if the pair edits are read in the incorrect order, then the pair edits should be combined. [13], [14], [18]

In the case of a contradictory pair, it is assumed that the pair contains information that cannot be reconciled without further justification or evidence. This includes pairs that contain claims that are in direct opposition to one another, pairs that contain claims that are edits that are temporally or numerically incompatible, or pairs that contain edits that are in direct opposition to an assumption that is required to be true by the other pair edit. CRDT convergence ensures that contradictions are preserved, but the content may violate an application or editorial requirement [6], [10], [13], [16]. These pairs are assumed to require human resolution.

An independent pair concerns different semantic targets or context regions and has no interaction beyond being simultaneous. Normally, both edits are kept without special justification. An exception is made at a high level of uncertainty or risk. [7]-[10]. The four labels used in this taxonomy are action-oriented. They each suggest the least intrusive next step to be taken rather than suggests edits directly. This taxonomy also does not consider every case of simultaneous modification as a conflict. Treating every case as a conflict would increase the workload on the reviewers and would make the semantically problematic cases harder to identify.

TABLE I. SEMANTIC RELATION TAXONOMY AND DEFAULT REVIEW ACTION

Relation	Operational definition	Default HITL action
Duplicate	Same semantic effect; little or no distinct information.	Auto-pass if confidence is very high; otherwise reviewer chooses one version or merged wording.
Complementary	Compatible additions that enrich the same topic.	Preserve both; optional ordering or combination suggestion.
Contradictory	Claims or editing intents cannot safely coexist.	Mandatory human verification before semantic reconciliation.
Independent	Different semantic targets; no meaningful interaction.	Preserve both; no human review unless uncertainty or risk is high.

Explanation: The taxonomy separates semantic compatibility from CRDT convergence and places each detected relation to the least restrictive action that guarantees safe collaborative editing.

B. Risk, Uncertainty, and Selective-Review Decision Rule

Selective review requires a decision rule that incorporates model uncertainty and the impact of an error. Let P be the four-class probability vector. Uncertainty can be defined as normalized entropy or equal to 1 minus the maximum class probability. The routing layer integrates uncertainty with the probability of contradiction, semantic-intention risk, and structural span overlap [12]-[16]. The Low, Medium, and High intent tiers are screening priors and evaluate the level of caution needed for different types of revisions; they are not precedent conflict labels.

A working formula is $R = w1U + w2C + w3I + w4O$, where U is uncertainty, C is contradiction probability, I is normalized intent risk, and O is structural span overlap. Before testing, the formula is validated and weights and thresholds are determined. They are not selected based on the test results or performance [13]-[16]. A pair is escalated to manual review if R exceeds the review threshold, contradiction probability crosses a mandatory-safety threshold, or there is a policy requirement to review a pair with a high-risk intent even if the confidence is low.

This decision rule is designed to facilitate selective prediction. Strict thresholds result in covering less pairs with the automated system, while more difficult pairs are referred to humans. This should result in a lower rate of semantic errors. On the other hand, more lenient thresholds increase the system coverage, but at the expense of more automated errors. The objective is to recognize the full trade-off between coverage and risk rather than highlighting a single, most favorable operating point [18]-[20]. Other review policies compare the review-all workload to the automated risk.

For the revisions here, tiers are recalculated directly from the codebook, so intent code 0 remains a valid high-priority category and is not combined with missing information [12]. The tier is used as an input feature and a stratification variable only. Semantic relations are established independently by human annotation and serve as authoritative evaluation labels.

V. EXPERIMENTAL METHODOLOGY

A. Corpus Preparation and Human-Validated Concurrent-Edit Benchmark

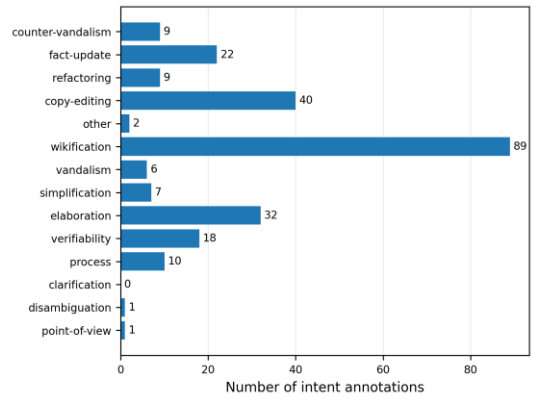
Benchmark construction begins with 300 Wikipedia revision records and 246 semantic-intent annotations from the revision-intent resource of Yang et al. [12]. Each record associates a revision with intent codes. The parent and revised text are extracted by the link to the revision or by the MediaWiki revisions interface, and local before-and-after spans and bounded surrounding context are extracted. Semantic-intent labels are used for stratified sampling and risk analysis, not for resolving pairwise conflicts.

Sampling keeps the observed variation of revision intents and keeps multi-intent cases, which may lead to more difficult semantic judgments. Twenty-six of the 300 revisions contain two or three intent annotations [12]. The review tier derived from the codebook is also retained to create High, Medium, Low, and Uncoded strata, and code 0 is treated as a valid counter-vandalism label. These strata are used to direct coverage of editing behaviors without predetermining the final relation class.

The existing records have no information on pairwise reviewer decisions, conflict types, CRDT resolutions, detection latencies, or reviewer notes. Therefore, concurrent-edit ground truth must be constructed separately for classification assessment. For each base passage sampled from the corpus, two candidate edits for the same or a nearby semantic target are constructed or retrieved and replayed from the same common state on two isolated copies of the system. Each edit pair is presented to two different annotators who evaluate whether the edits are duplicate, complement, conflict, or are independent of each other. A tie is broken by a third reviewer. Cohen's kappa is reported for paired annotations, and Fleiss' kappa is reported for three-way annotations. [13], [16], [18]. Article or base-passage identifiers are kept so that subsequent evaluation splits

can avoid closely related edits from being assigned to the training and test sets.

Fig. 2. Distribution of Semantic Edit-Intent Annotations



Source: Counts computed from the Wikipedia revision-intention records described by Yang et al. [12].

Explanation: Wikification makes up the lion's share of labeled edits, with copy-editing and elaboration making up the next largest groups, resulting in a naturally imbalanced semantic-intent evaluation set.

B. Baselines, Controlled Offline-First Replay, and Ablations

Five systems are assessed against the same candidate pairs. B0 is a baseline using structural rules with target, lexical, and distance overlap. B1 uses Sentence-BERT with calibrated class thresholds [15]. B2 uses a natural language inference classifier with the generative transformer [13], [14]. B3 uses a prompt-based large language model [19], [20]. B4 is the proposed system that combines structural pruning, embeddings, bidirectional inference, large-model escalation, and human review for uncertain/high-risk cases.

Each labeled pair is run through the Yjs editor with two clients branched from a common document state. Connectivity is turned off before edits are made, replicas perform their local updates, and connectivity is then turned on. Semantic scoring is performed once the Yjs replicas have synchronized and convergence is achieved. This is to avoid classification performance being affected by replication failures [5], [7]-[10]. Replays capture candidate metadata, the timestamps for each phase, model invocations, routing decisions, and reviewer actions.

Ablation experiments remove one component at a time: They differ in intent-risk prior, structural candidate pruning, bidirectional inference, large-model escalation, and human-review threshold. Measuring these differences tells us which stages significantly impact accuracy, contradiction recall, efficiency, and review savings [12]-[16], [19], [20]. To address leakage, we will evaluate on articles or base passages rather than on random sampling of edit pairs. Repeated stratified folds or a fixed train, validation, and test partition may be used, with any class balancing restricted to the training data [13], [14], [18]. Threshold tuning and calibration are done on the validation data rather than the test data to ensure consistent evaluation. Finally, evaluation is done on the test data to get a final score.

C. Evaluation Metrics and Statistical Analysis

Evaluation distinguishes semantic quality, human-review burden, and systems overhead. The primary semantic measure is macro-F1 across the four relations, along with Precision and Recall for each class [13], [14]. We report Contradiction Recall to capture the case where a contradiction is missed, resulting in a converged document containing contradictory claims. We assume that an escalation is unnecessary only if the document to be reviewed is contradiction-free, and otherwise, it is a burden on the reviewer. We inspect confusion matrices to distinguish between different types of errors to help classify contradictory errors, complementary errors, duplicate errors, and independent errors. We present the results based on semantic-intent and review-risk tier [12]-[16].

Examples of human-in-the-loop metrics are the percent of candidate pairs escalated, the review-workload reduction compared to a review-all policy, the time spent by reviewers per 100 edits, the

correction rate of automated decisions, and inter-annotator agreement [17], [18]. Automated coverage is the proportion of candidates resolved without any human intervention. LLM-avoidance coverage is the proportion of candidates where an LLM is not used. The result reported as 80.7% in Section VI uses LLM-avoidance coverage, not no-human-intervention coverage. Coverage is reported with residual error or contradiction risk to ensure that an overall low review rate is not misinterpreted as a benefit, even if it is achieved by accepting risky cases.

Examples of systems metrics are the median and 95th percentile of additional detection latency, large-model calls per 100 candidates, tokens and cost per pair if applicable, memory overhead, and CRDT convergence success [7]-[10], [19], [20]. The last metric ensures that the semantic observer has not modified the replicated state. For statistical comparisons, we use paired bootstrap confidence intervals for macro-F1, and paired classification-error differences for a McNemar test [13], [14]. We report effect sizes and confidence intervals along with p-values. We set thresholds, use specific model versions, prompts, and random seeds to ensure that all metrics can be reproduced, and to evaluate the sensitivity to changes in configuration.

VI. PRELIMINARY RESULTS AND EVALUATION PROTOCOL

A. Preliminary Corpus Audit and Descriptive Findings

This preliminary audit synthesizes the empirical evidence from the corpus together with the completed full-cascade evaluation reported below. The corpus comprises 300 revision records and 246 semantic-intent annotations. Of the 300 revisions, 215 contained at least one intent code, and 85 contained none. Consequently, 215 or 71.7% of the revisions can provide an intent-based prior. Annotation multiplicity is evident across the records. Of the 300 revisions, 189 contained a single annotation, 21 contained two annotations, and five contained three annotations. Therefore, 8.7% of the records, or 26 revisions, have multiple intents. These multi-intent cases provide a more challenging and informative stratum because one revision can represent multiple editorial motivations.

There is a heavy imbalance in the frequencies of the intended edits. Wikification was the most prevalent category with 89 annotations, followed by copy-editing with 40 and elaboration with 32. Fact-update and verifiability edited a total of 22 and 18 annotations, respectively. Less prevalent edits included process, counter-vandalism, refactoring, simplification, and vandalism, while clarification had no annotations. Disambiguation and point-of-view edits each appeared once. Given this edit distribution, it would be more prudent to employ macro-averaged reporting and stratified sampling, as opposed to relying on the overall accuracy of the data, which could be skewed by the high prevalence of low-risk edits. [12].

A different quality check finds nine revisions with Intent_Codes value 0 and the stored human-review tier set to Uncoded. The codebook states that 0 represents counter-vandalism and sets it as a High priority code. If we take the original records and recalculate the review tier using the codebook, the totals for the tiers change from High 47, Medium 41, Low 118, and Uncoded 94 to High 56, Medium 41, Low 118, and Uncoded 85. The nine inconsistencies are therefore a coding-interpretation issue, not a semantic conflict.

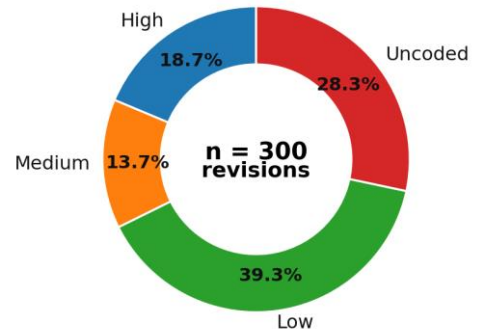
Finally, the values for Reviewer_Decision, Conflict_Type, CRDT_Resolution, Detection_Latency_ms, and Reviewer_Notes are all empty for the 300 revisions. These values cannot influence any of the classification metrics (accuracy, recall, etc.) or latency, and therefore indicate the need for a human-validated pairwise comparison and controlled replay as described in Section V, prior to any model comparison. We can interpret the lack of data in these fields as a positive indicator (rather than a quality failure), because the corpus was originally assembled for semantic-intent labeling, not concurrent-edit conflict evaluation. Their emptiness demonstrates that pairwise judgments, reconciliation logs and timing data must be obtained via specialized annotation/replay so that the descriptive audit and the classification evaluation remain separate.

TABLE II. PRELIMINARY AUDIT OF THE 300-REVISION CORPUS

Audit item	Observed / derived value	Use in paper
Revisions	300	Seed corpus size
Intent annotations	246	Intent distribution / stratification
Coded revisions	215 (71.7%)	Eligible intent-prior records
Multi-intent revisions	26 (8.7%)	Harder multi-label stratum
Raw HITL tiers	H 47 / M 41 / L 118 / U 94	Original recorded values
Derived HITL tiers	H 56 / M 41 / L 118 / U 85	Codebook-consistent analysis
Tier mismatches	9 code-0 records	Data-quality finding
Conflict / decision / latency labels	0 populated of 300	Requires human annotation and experiments

Explanation: The audit keeps the originally recorded values and uses codebook-derived tiers for analysis, allowing us to distinguish the nine code-zero records from being treated as unlabeled.

Fig. 3. Codebook-Derived HITL Review-Tier Distribution



Source: Review tiers recomputed from the revision-intention codebook associated with Yang et al. [12].

Explanation: Recalculating review tiers from the codebook shows that 56 records are high-priority. This includes nine counter-vandalism records that were previously stored as uncoded.

B. Comparative Evaluation Status and Reporting Protocol

The source records do not contain the pairwise relation labels or measured replay outputs needed for a complete B0-B4 comparison. Accordingly, the present paper does not report a complete baseline comparison or replay-derived latency statistics for B0-B3. The completed full-cascade evaluation of B4 on 300 independently labeled edit pairs is reported below, while broader baseline and replay comparisons remain future work. This distinction reflects the difference between semantic-intent labels, which annotate individual revisions [12], and pairwise inference labels, which describe duplication, complementarity, contradiction, and independence [13], [14], [18].

Cohen's κ is 0.800 and raw agreement is at 85%. There are 45 disputed pairs which a third reviewer settled. A thorough auditing of 300 pairs of edits yielded an LLM coverage rate of 80.7%. 242 out of 300 pairs (80.7%) were safe to handle without intervention from the LLM, while 58 pairs (19.3%) required escalation to an LLM. Testing the cascade for its ability to detect true contradictions was also promising with an 87.3% (69 out of 79) success rate. From this, we might conclude that the cascade trades off contradiction precision for high recall on the current benchmark. The complete cascade was also demonstrated to have balanced performance on all four relations with a high overall accuracy of 75.7%, macro-F1 score of 0.753, and a recall rate of 53% ($F1 = 0.60$) for the complementary class. The 19.3% pairs escalated to an LLM probably represent actual semantic ambiguity as opposed to low-cost resolution, indicating a need to sacrifice coverage for precision; this is a subject for future work. As mentioned above, this pattern is consistent with the routing philosophy described at the beginning, which seeks to direct reviewer efforts to higher risk pairs while keeping contradiction recall high.

TABLE III. FULL-CASCADE RESULTS (N=300; $\kappa=0.800$ inter-annotator agreement). Accuracy=75.7%, Macro-F1=0.753, LLM-Avoidance Coverage=80.7%, Contradiction Recall=0.873.

Class	Precision	Recall	F1
Duplicate	0.79	0.74	0.76
Complementary	0.69	0.53	0.60
Contradictory	0.73	0.87	0.79
Independent	0.82	0.89	0.85

In future comparative work, B0 through B4 should be evaluated on the same test pairs using macro-F1, class-specific F1 scores, contradiction recall, automated coverage, review rate, median and p95 added latency, and large-model calls per 100 candidates [13]-[20]. The proposed cascade should be considered better only when semantic improvements and review savings justify the additional stages. A lower review rate alone is not adequate evidence of improvement, because an aggressive threshold could save reviews by silently accepting contradictory edits. A large-model-only baseline could also achieve strong semantic scores while introducing unacceptable latency and cost. The comparison should therefore consider quality, review burden, and systems overhead together.

Future comparative evaluation should use coverage-risk curves rather than a single threshold to characterize selective-review behavior. Each point should show the percentage of candidates processed automatically and the corresponding risk of a semantic error or contradiction. Operating points should be selected without tuning on the test set used for final evaluation [18]-[20]. Confusion matrices should distinguish errors between complementary and duplicate classes from errors involving contradictory classes. Errors should also be examined across High, Medium, and Low review tiers to assess whether risk-aware routing protects significant edits [12]-[16].

There are four main considerations for interpreting the present results and future comparisons. First, the 300-pair evaluation provides an initial estimate of B4 performance but does not establish superiority over B0-B3. Second, selective review should be compared with review-all to quantify workload reduction. Third, the intent-risk prior should be tested for contradiction recall in high-priority cases. Fourth, semantic triage is designed to preserve Yjs convergence while maintaining bounded additional latency [7]-[10], [12]-[20]. Qualitative examples are not used to justify the present quantitative results; future comparative work should include representative contradiction, complementary, duplicate, and error cases to illustrate decision boundaries [17]-[20]. Future comparisons should also report confidence intervals, paired significance tests, and replay seeds so readers can distinguish sampling variability from systematic improvements and reproduce the operating points.

VII. DISCUSSION

The main advancement is an integration of existing lines of work. This includes offline-first CRDT-based collaborative rich-text editing, semantic analysis of simultaneously generated editing operations, and human-in-the-loop reconciliation. This integration addresses the separation of concerns of replicated state correctness and content-level editorial correctness. CRDTs and local-first approaches ensure that editing operations are preserved, support working on the editing document without being connected to the network, and bring replica editing documents to a common state. [5], [7]-[9]. The semantic layer is concerned with concurrent editing spans and considers a different question: are the remaining edits in natural language contradictory? This question reframes Yjs convergence semantics to ensure language model uncertainty does not propagate to the replication protocol. This is also congruent with the integrity research that indicates that from a technical perspective, merging two documents is always possible, but that does not guarantee that the documents can be merged from a semantic perspective. [6]

Automatic generative rewriting poses a greater threat to the integrity of collaborative documents, as disputed content is generated by human collaborators. Compared to selective human verification, interpreting an ambiguous document pair with a large language model is less computationally expensive and may be less harmful,

though it could make edits that remove attribution. The cascade ensures that the greatest uncertainty and potential harm are addressed by spending the most computation and reviewer time on those cases. Clear, independent, and complementary cases are allowed to proceed automatically. Editing suggestions may be automatically generated for cases that likely contain duplicate editing operations, and the remaining cases that contain editing operations of an uncertain nature and high risk are left for review. Automatic handling systems are limited to adding preservable high-certainty independent and complementary changes, as well as nondestructive suggestions for high-certainty duplicates. Contradictory, uncertain, or high-risk changes will remain under review. Treating human review as a scarce resource avoids providing automation with decision-making authority over disputed meaning.

The four-class taxonomy generates more informative product actions than a binary conflict flag. Duplicate and complementary edits are semantically related but require different types of resolutions. The same is true for contradictory and independent edits, but only contradictory edits require reconciliation. Therefore, remembering contradictions is more essential for safety and quality than overall accuracy, which can easily be improved by including frequent, simple classes [13], [14], [16]. This distinction supplements semantic conflict modeling and revision intention analysis at the operation level [10], [12]: causal metadata indicates when edits conflict, intent labels evaluate the risk of a review, and the relation is determined by NLP.

The types of revision observed in Wikipedia represent a wide variety of editorial behaviors, but their frequency does not necessarily represent other collaborative environments. Longer, more rigid dependence sequences can be found in enterprise notes, software requirements, policy documents, and scientific papers, along with stronger terminological constraints and different implications for factual disagreements [12], [18]. Therefore, it is necessary to adapt the protocol for other domains before making broad claims about the reduction of incremental work or threshold adjustments. It can be carefully readjusted for different writing domains.

VIII. LIMITATIONS

There are several limitations on the conclusions that can be drawn at this stage. First, the 300 revision records were collected for semantic edit-intent analysis, not for concurrent-conflict classification. They contain no pairwise reviewer decisions, relation labels, reconciliation outcomes, or detection latency measurements. A separate human-validated pairing and annotation procedure is therefore required to assess semantic accuracy, contradiction recall, and reviewer workload [12], [18]. The corpus audit supports characterization and risk stratification, while the separate 300-pair full-cascade evaluation provides a substantive performance assessment. A full cascade evaluation (N=300, two-annotator $\kappa=0.800$) showed that lightweight filters still favor contradiction sensitivity over precision on the complementary and duplicate classes. This bias serves safety (high contradiction recall). However, it shows that the current similarity and entailment thresholds could be further adjusted, and the cascade's precision claims should continue to be validated at larger scale.

Second, while controlled offline replay increases the repeatability of the experiment, it will never fully recreate the social and contextual features of collaborative work. Real world teams may conduct change discussions that span beyond the editor, or revise over long time intervals. These can even influence whether statements that appear to be inconsistent are in fact contradictory [5], [17], [18]. To gauge the validity of the real world case, more deployment studies will need to be conducted.

Third, there can be large variance in judgements across different versions of large models, as well as different prompts and sampling. Additionally, the rationales generated by the model can be incorrect and lack reasoning [19], [20]. For the evaluation, it will be necessary to document the identifiers of the models used, as well as the prompts, and use deterministic settings. For the human labels, it will be considered ground truth, and model explanations will be disregarded.

Finally, structural pruning is used to reduce the amount of work the system needs to do, and can favor more local, paragraph level interactions. This can lead to semantic contradictions being ignored, such as a revised definition that invalidates a claim that was made several sections prior [10], [18]. The initial focus is on a more localized scale. Longer document-level dependency tracking, retrieval, and contradiction chains are proposed for future work and are not evaluated within the present study. These limitations will be used to justify a more conservative interpretation and more transparency in reporting before broader deployment claims for collaborative work will be made.

IX. CONCLUSION AND FUTURE WORK

Offline-first CRDT editors can preserve concurrent work and converge after reconnection while still producing prose that requires semantic judgment. The framework presented here focuses on the post-convergence layer and does not modify CRDT merge rules. As part of the integration of offline-first CRDTs, semantic analysis of concurrent edits, and human-in-the-loop reconciliation, this work makes no novel claim on semantic conflict detection. It uses a combination of concurrency and locality metadata, lexical features, Sentence-BERT, bidirectional NLI, selective LLM escalation, and human-in-the-loop edits to classify pairs of edits as duplicate, complementary, contradictory, or independent [5], [7]-[10], [13]-[20]. Each of these edit classifications inspires a conservative action to be taken to ensure that editing collaborators retain control.

The empirical evidence supports both corpus-level descriptive findings and an initial full-cascade performance assessment, while broader baseline comparisons remain future work. An audit of 300 Wikipedia-derived revisions contains 246 intent annotations, 215 coded revisions, 26 revisions containing multiple intent codes, and nine tier 0 code-mismatch cases. Recomputing tiers from the codebook resulted in 56 High, 41 Medium, 118 Low, and 85 Uncoded records. Conducting a full-cascade evaluation on 300 two-annotator pairs ($\kappa = 0.800$) shows great strides in LLM-avoidance coverage (80.7%) and recall for contradictions (87.3%) and confirms the safety-critical routing design. A classification accuracy of 75.7% shows balanced performance across the four relations, with the complementary class reaching 53% recall. Continued threshold recalibration to further reduce contradiction over-estimation remains a target for the next phase of this work. The next step is to expand the human-validated concurrent-edit benchmark, complete baseline comparisons, and extend offline/reconnection replay. Future reporting should focus on class-specific precision, recall, and F1 scores; selective-review coverage and workload; latency and model-call costs; and whether semantic monitoring affects Yjs convergence. The goal of future work will be to move from local edit pairs to document-level contradiction chains, provide more evidence for fact updates, utilize adaptive thresholds and privacy-preserving semantic models, and perform on-device semantic modeling, as well as field studies with real collaborative teams. These improvements will help evaluate if the divide between replication and semantic review will still hold true in more extensive domains and longer documents, as well as in different deployment circumstances.

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